

Profile

Mirror Image Internet, Inc.

49 Dragon Court
Woburn, MA 01801
(800) 353-2923
www.mirror-image.com

Mirror Image Internet Advances Delivery Services for Current and New Customers

Preface

Customers engage content delivery network (CDN) service providers for two purposes: to increase the speed of end-user access to Web pages and to cut the costs of Web site delivery. The number of CDNs peaked in 2001, when Aberdeen tracked more than 40 vendors. The stringent market challenges of the last three years pared the field to fewer than a dozen.

However, usage of CDNs rose dramatically during that same period. Today, most corporations, governments, and educational institutions rely on CDNs to increase the availability and reliability of public-facing Web sites and to economically circumvent the unknowns and inefficiencies of the open Internet. CDNs are standard procedure for delivering most types of Web-accessed content, from static images and streaming media to whole sites and downloadable software.

The CDN suppliers left standing are all the stronger for surviving the dot-bomb trials. They are rapidly developing new services to move beyond basic image, page, and media delivery. The objective is to enable customers to deliver applications over the Internet as reliably as they deliver static and streaming content. Leveraging their core competence of building and maintaining Internet overlay applications to meet these demands, CDNs are combining existing services with new technologies to offer comprehensive application delivery solutions.

Executive Summary

Mirror Image Internet (Mirror Image) is a CDN service provider that has weathered the storm and emerged as one of the top CDN vendors. Aberdeen interviews with Mirror Image customers, reflected in two case studies included in this *Profile*, show that the supplier admirably meets the availability and cost-cutting require-

ments for CDNs. Customers rate the company highly on responsiveness, reach, and customer service.

Building on its core content delivery services, Mirror Image offers a suite of “Smart Content” services to empower non-information technology (IT) groups — such as customer service, marketing, and e-commerce — with direct control over delivery of their content. These services reduce the pressure on IT and Web operations to mediate the content flow between content creation groups and external users.

Mirror Image Internet

Founded in 1997, Mirror Image is a wholly owned subsidiary of Xcelera.com, a publicly traded Internet holding company (Amex: XLA). Mirror Image is headquartered in Woburn, MA, and has 75 employees.

Mirror Image’s approach to CDN services is founded on its Content Access Point (CAP) architecture that combines global delivery with centralized control. CAPs are specially designed data centers located at critical regional Internet crossroads throughout the world. Mirror Image moves content to the CAPs, from which end-users access the information.

Mirror Image’s approach to CDN services is founded on its Content Access Point (CAP) architecture that combines global delivery with centralized control. CAPs are specially designed data centers located at critical regional Internet crossroads throughout the world

In today’s Internet, with subsecond responses within regions, the CAP approach proves two core benefits. First, the regional architecture minimizes Mirror Image’s infrastructure costs, which translates into lower client costs. Second, the CAP data centers increase the control, security, and manageability of the customer’s content in the network.

The CAPs also allow customers to leverage the processing power of Mirror Image’s dis-

tributed data centers, which are different than just a set of caches. Current customers include Global 2000 enterprises, e-businesses, and service and hosting providers. Coming from vertical markets such as retail, advertising, and print and broadcast media, Mirror Image’s customers sign up for the following types of services:

- *“Smart” Content Delivery* — Global content caching, digital download, geotargeting, business analytics, and business continuity capabilities to accelerate end-user access and cut customers’ bandwidth and infrastructure costs.
- *Delivery Reporting and Content Control* — Near-real-time bandwidth and transaction reporting and granular control of every piece of content

throughout the network enable customers to understand actual usage quickly and accurately.

- *Streaming Media* — By including on-demand and live Web casting options, Mirror Image enables customer communication with employees, partners, and customers via Internet-based audio and video content.

Customers Attest to Mirror Image's Value

Mirror Image has gathered a fleet of loyal customers over the past six years. Aberdeen continually contacts customers to understand their ongoing and emerging needs for content distribution. The following two case studies reflect the value that Mirror Image provides to its current customers, as evidenced by Aberdeen research, and indicates what customers have told Aberdeen that they want to see in the future.

Fastclick, Inc.

Much like the content delivery industry, online advertising was hit hard during the economic downturn of 2000 through 2002. Founded and funded just months before the market dropped, Fastclick survived owing to (a) good financial management and (b) an advanced technology platform designed to accommodate growth and manage expenses for Internet advertising services. In late 2003, Fastclick was delivering more than 5 billion impressions per month using two services:

- *Ad Network*, which connects 14,000 buyers and sellers of online ad inventory
- *AdServer*, a high-performance engine that hosts and serves Internet advertising and includes Web-based tools for customers to correlate pricing models with advertising campaigns and build positive ROIs

Steve Chang, the company's chief technology officer, reported that there are three technology demands for serving the high volume: hardware, bandwidth, and security. As the company scaled up in mid-2002, Chang recognized that these three demands would become bottlenecks to ad-serving performance and company profitability. Scaling at least one of the three was required to meet the growing need, but doing so would eat into the profit margins. Building out all three areas was completely infeasible given the budget constraints.

Chang decided to use a CDN service provider to "reduce bandwidth and infrastructure costs." "With a CDN, we needed fewer servers, less room, less power, and less floor space to meet our performance thresholds," he said. Chang chose Mirror Image because "price was a huge issue," and Mirror Image was willing to work with Fastclick to build a pricing model that "took into account [its] growth trajectory over time." Mirror Image was also willing to "go the extra mile" and form a tight business and technology relationship.

Although Fastclick's reporting requirements are straightforward — as Chang said, "All I really need to know is how many ads did we serve?" — he also acknowledged that engaging Mirror Image averted many security concerns. "I don't have to worry about traffic filtering or intrusion detection because everything goes through the Mirror Image network and never hits our data center directly."

"I don't have to worry about traffic filtering or intrusion detection because everything goes through the Mirror Image network and never hits our data center directly."

— Steve Chang, chief technology officer at Fastclick, Inc.

One of the largest benefits for Chang was simply time. By not having to focus on scaling the bandwidth, server, or security issues, his team could enhance Fastclick's products faster and increase the services for his customers. "Mirror Image has a specific function for us, and we're very pleased that [it] performs that function particularly well," Chang said. Currently, Chang is working with Mirror Image on new technology and services — in particular, how to add the "adaptive content" action rules to improve

geotargeting and ad rotation. Mirror Image continues to address the unique needs of Fastclick, Poindexer, Pointroll, and other online advertisers with a focused service package for the vertical market.

An International Toy Retailer

Aberdeen interviewed the director of Internet operations at a major international toy retailer (who requested anonymity), a longtime Mirror Image customer. The director has been with the company for 25 years and has overseen Web operations for the last six years. He oversees the company's infrastructure, hosting, and Web application design and operation.

From the launch of its online site, the company's goal for its Web operations was to make customers "more comfortable with our products, which leads to a more easy transition to shopping." The e-commerce elements of the site are "as close to a brick-and-mortar store as possible," with the products organized similar to a physical store and using the same layout as the company's catalogs. A priority was also set for the company's online operations to develop a reputation that mirrored the high quality associated with its worldwide brand.

Though e-commerce is certainly part of the site, much of the site is not sales related. Consumers are encouraged to sign up for free services that tie online games, puzzles, and activities to physical products. The current enrollment is "in the millions of members." The company continually creates new interactive components, movie clips, storyboards, and downloads to encourage children and parents to use the company's products as creatively and regularly as possible.

The company hosts the Web applications at two data centers, one in North America and one in Europe. However, the director of Internet operations needed to serve content to millions of users worldwide in their native languages. He said, “I wanted two things: flexibility in operations to deliver more content with minimal additional cost and fast performance by delivering the content from as close to the user as possible.” The director understood that he could build data centers in every major worldwide region, but deploying them would be expensive, and managing them would be difficult. As he said, “I already spend enough time on planes. I really don’t want to have to travel everywhere in the world.”

When shopping for CDNs, the director looked at the three major vendors available at the time. All proved quickly that they could accelerate the site, offload pressure from the data centers, and reach consumers throughout the world using the two data centers as origins. The quoted pricing was also similar.

The director chose Mirror Image because of its “centralized yet distributed” architecture, ease of implementation, high flexibility, and strong reporting. “Mirror Image did not ask us to adapt our site to [its] options. [Mirror Image was] flexible and worked with us to customize the service for our needs.” As a result, Mirror Image “improved *significantly* the site’s performance,” and he is “very pleased that he can scale very cost-effectively.”

The director also stated that Mirror Image was “incredibly responsive during and after the sales process, and in responding to requests for new functionality.” To illustrate this point, he said, “I asked Mirror Image for a reporting tool that would let me monitor, in real time, my content throughout [Mirror Image’s] network. Mirror Image moved from my request to a production level tool in less than 45 days. The resulting tool lets me see my content throughout the Mirror Image network and also validates the SLAs [service level agreements]. Now, my staff can talk the same language as MI’s staff. When there is a problem, we understand immediately whether it’s our network or Mirror Image’s.”

Adding to the comments about cooperation, the director said that his team works closely with Mirror Image technicians. “The MI staff is almost a part of my staff,” he said. “MI’s support team helps resolve any problems that come up, whether or not the problems are from Mirror Image.”

The director is very interested in Mirror Image’s next service offerings. For a retailer, “adaptive caching is central to making on-the-fly content decisions based on how specific content is used in particular regions.” Although he readily admits that his company is conservative in its adoption of new technologies, he said that he will watch closely how other Mirror Image customers use the adaptive content services as he plans the next evolution of his architecture in preparation for the next holiday season.

New Services from Tested Building Blocks

Responding to customers' needs, Mirror Image has aggregated its technology and service elements into its "Smart Content" services platform. This platform leverages elements of the CAP architecture and marries them with an advanced XML-based rules engine and enhanced report tools, such as those developed for the toy retailer.

Each customer will be able to program its own content rules into the engine, based on geotargeting, time of day, place in rotation, "wild card" image replacements, and other parameters. These rules are distributed with the content throughout the global network. End-users are served from a local data center, using the XML rules and the locally cached content.

Responding to customers' needs, Mirror Image has aggregated its technology and service elements into its "Smart Content" services platform. This platform leverages elements of the CAP architecture and marries them with an advanced XML-based rules engine and enhanced report tools, such as those developed for the toy retailer.

Mirror Image expects IT, Web operations, and other technical groups to benefit from the Smart Content platform in two ways. First, the new services allow greater visibility into actual CDN usage to confirm the benefits of speed, efficiency, and accuracy. Second, the reporting and user interfaces will allow nontechnical groups to manage the flow of content in, out, and around the CDN. Once IT or Web operations install the distribution rules, the customer's marketing group, e-commerce team, or other non-IT staff can insert content directly into Mirror Image's network without involving IT or Web operations.

As a result, IT and Web operations staff will be able to remove themselves from being the key masters of the content delivery services. Customers such as Fastclick's Steve Chang will have more time to focus on nondelivery issues. The director of Internet operations at the toy retailer will be able to develop XML rules that customize content to local audiences and let the marketing department own that customization process.

Aberdeen Conclusions

For many years, CDN and caching suppliers have promised to separate managing the content in the network from managing the network itself. Mirror Image customers attest to the need for greater visibility into where their specific content is accessed in the supplier's network. Whereas CDNs can increase flexibility, improve response times for end-users, and lower the scalability costs for Web operations, customers of CDNs need verifiable data before they will shift more content and application delivery responsibility to the CDNs.

The Smart Content platform continues to provide the core CDN values while increasing visibility and flexibility of content delivery and consumption. Aberdeen research with Mirror Image customers verified that the expected benefits are being achieved. That is no small task, given the tumultuous climate of the past three years. The challenge facing Mirror Image — and all CDNs — is how to increase its value to customers. The baseline services of straight image and Web site delivery are under significant price pressure.

The new services take advantage of the CAP design feature of Mirror Image's network, which centralizes processing power while also positioning content for local local access in major geographic regions. Thus, customers get fast end-user delivery and access to local processing power, two major steps toward distributed application delivery. Aberdeen recommends that IT groups at customer organizations — in conjunction with their marketing, e-commerce, and creative design teams — should evaluate these capabilities. The next stage of Web application delivery evolution will benefit greatly from central management of rules that are applied on a distributed basis.

The Smart Content platform continues to provide the core CDN values while increasing visibility and flexibility of content delivery and consumption. Aberdeen research with Mirror Image customers verified that the expected benefits are being achieved.

To provide us with your feedback on this research, please go to www.aberdeen.com/feedback.

*Aberdeen Group, Inc.
260 Franklin Street
Boston, Massachusetts
02110-3112
USA*

*Telephone: 617 723 7890
Fax: 617 723 7897
www.aberdeen.com*

*© 2004 Aberdeen Group, Inc.
All rights reserved
May 2004*

Founded in 1988, Aberdeen Group is the trusted advisor to the Global 5000 for value chain strategies and technology advice. Through its continued benchmarking and analysis of value chain practices, Aberdeen offers a unique mix of research, tools, and services to help G5000 executives assess their value chain performance, develop improvement strategies, and select value chain solution partners.

This document is the result of research performed by Aberdeen Group that was underwritten by Mirror Image. Aberdeen Group believes its findings are objective and represent the best analysis available at the time of publication.